1. Tail control

For independent $Z_1, \ldots, Z_n$, write $\hat{\mu}_n = n^{-1} \sum_i Z_i$ and target $P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \delta$. State assumptions before the bound.

1. Markov and Chebyshev

Markov: if $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq \frac{E[X]}{a}$.

Chebyshev: if $E[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

For i.i.d. variables,

$$\mathrm{Var}(\hat{\mu}_n) = \frac{\sigma^2}{n}, \quad P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

The variance reduction uses independence.

2. Hoeffding inequality

If independent X_i satisfy $a_i \leq X_i \leq b_i$ almost surely, then

$$P\left(\sum_{i=1}^{n(X_i - E[X_i])} \geq t\right) \leq \exp\left(-2\frac{t^2}{\sum_{i=1}^{n(b_i - a_i)^2}}\right).$$

Apply it to both signs for a two-sided bound. For independent $Z_i \in [0, 1]$,

$$P(|\hat{\mu}_n - \mu| \geq \varepsilon) \leq 2\exp(-2n\varepsilon^2), \quad n \geq \frac{\ln\left(\frac{2}{\delta}\right)}{2\varepsilon^2}.$$

Use Markov with a nonnegative variable and its mean, Chebyshev with finite variance, and Hoeffding with known bounds plus independence. Bounded dependent time-series losses do not justify this Hoeffding form.

2. Vector geometry

1. L^p norms

For $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$ and $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{j=1}^{d} |x_j|^p\right)^{\frac{1}{p}}, \quad \|x\|_\infty = \max_j |x_j|.$$

If $1 \leq p \leq q \leq \infty$,

$$\|x\|_q \leq \|x\|_p \leq d^{\frac{1}{p} - \frac{1}{q}} \|x\|_q.$$

Fixed-dimensional norms induce the same convergence, but their constants and level-set shapes are not interchangeable. For $0 < p < 1$, the expression can fail the triangle inequality and is not called a norm here.

Changing the norm can change the optimizer's structure: maximizing $2x_1 + x_2$ over the L^1 unit ball gives sparse $(1, 0)$, while over the L^2 unit ball it gives $\frac{2,1}{\sqrt{5}}$.

2. Inner product and Cauchy–Schwarz

For $x, y \in \mathbb{R}^d$,

$$\langle x, y \rangle = \sum_j x_j y_j, \quad \|x\|_2 = \sqrt{\langle x, x \rangle},$$

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

Equality holds exactly when x and y are linearly dependent, including the zero-vector cases. For $y \neq 0$, substitute $t = \langle x, y \frac{\rangle}{\|y\|_2^2}$ into $0 \leq \|x - ty\|_2^2$.

3. Singular-direction geometry

1. Compact singular-value decomposition

If $A \in \mathbb{R}^{m \times d}$ has rank r,

$$A = U_r \Sigma_r V_r^T, \quad \Sigma_r = \mathrm{diag}(\sigma_1, \ldots, \sigma_r), \quad \sigma_1 \geq \ldots \geq \sigma_r > 0,$$

where U_r, V_r have orthonormal columns. For each positive pair,

$$A v_j = \sigma_j u_j, \quad A^T u_j = \sigma_j v_j.$$

v_j is an input direction, u_j the matching output direction, and σ_j the stretch. The compact form contains only positive singular pairs.

2. Range, kernel, and inverse behavior

$$\mathrm{range}(A) = \mathrm{span}\{u_1, \ldots, u_r\}, \quad \mathrm{range}(A^T) = \mathrm{span}\{v_1, \ldots, v_r\},$$

$$\ker(A) = \mathrm{span}\,\{v_1, \ldots, v_r\}^\perp, \quad \mathrm{rank}(A) = r.$$

$$A^\dagger = V_r \Sigma_r^{-1} U_r^T, \quad A^\dagger u_j = \frac{1}{\sigma_j} v_j, \quad \|A\|_2 = \sigma_1.$$

$A^\dagger$ does not recover a kernel component. A zero singular value erases a direction; a small positive one is invertible but amplifies perturbations by $\frac{1}{\sigma_j}$.

4. Functions as vectors

1. $L^2([0, 1])$ and Hilbert spaces

$L^2([0, 1])$ contains real functions with $\int_0^1 |f(t)|^2 \, dt < \infty$, identifying functions equal almost everywhere. Define

$$\langle f, g \rangle = \int_0^1 f(t) g(t) \, dt, \quad \|f\|_2 = \sqrt{\int_0^1 |f(t)|^2 \, dt}.$$

A **Hilbert space** is an inner-product space complete in its induced norm: every norm-Cauchy sequence converges in norm to an element still in the space. $L^2([0, 1])$ is Hilbert, and Cauchy–Schwarz holds with the integral inner product.